Target validation paths — gist-sdh-multifocal-resected-m1-k4n8

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

The case turns on two tests. The SDHB IHC re-cut (clone 21A11 or BSB-131) paired with an SDHA IHC (clone 2E3 or D6J9M) resolves the SDHB-intact / biallelic-SDHA paradox, and the germline SDHA panel settles inheritance and family risk. Together they confirm the SDH-deficient driver that gates the belzutifan wt-GIST cohort via NCT04924075 and the temozolomide trials NCT03556384 and NCT05661643, and the germline result triggers cascade testing of first-degree relatives. If SDHA IHC also reads intact after a clean re-stain and the germline panel is unrevealing, the dSDH label collapses: there are no within-scope trial routes, SDHC promoter methylation becomes the next call, and standard KIT/PDGFRA-WT care for first recurrence is the treating team's conversation, not this report's.

SDH-deficient GIST

Three workups are essential before any feature-targeting therapy can be chosen. The germline SDHA full-gene sequencing panel with del/dup analysis resolves whether SDHA L306P (VAF 44%) is germline or fully somatic; tumor-only NGS cannot distinguish the two. The SDHB IHC re-cut with a positive internal control addresses the most likely explanation for retained SDHB, a technical false-negative from under-fixation or a missing internal-control vessel. The paired SDHA IHC arbitrates the conflict more directly, since SDHA staining is lost only in biallelic-SDHA tumors and retained when an SDHB/SDHC/SDHD or SDHC-methylation driver is at work. Two further essentials are patient-safety floors for an SDHx carrier: baseline paraganglioma surveillance (rapid-sequence whole-body MRI skull base to pelvis, plus Ga-68 DOTATATE PET/CT as the most sensitive functional tracer) and plasma free metanephrines by LC-MS/MS, because an undetected functional paraganglioma going to anesthesia is a hypertensive-crisis hazard. High priority but not gating: tumor-tissue NGS for SDHA E350fs, which maps clonal architecture (biallelic inactivation is already established by L306P plus Y408N), and tumor-informed ctDNA MRD (Signatera) as an imaging adjunct that is informative rather than therapy-triggering. Medium priority as fallbacks: SDHC promoter methylation if the SDHA story collapses (academic referral, 4 to 8 weeks), and MGMT promoter methylation, which predicts temozolomide response without gating enrollment and can wait until that agent enters the discussion.

PIK3CA R93W

Tumor-tissue NGS confirmation on a comprehensive panel (FoundationOne CDx, Tempus xT, or equivalent) is essential before the PI3Kα pathway can be acted on. R93W sits outside the validated alpelisib companion-diagnostic hotspots (E542K, E545K, H1047R, C420R) and the call is ctDNA-only, so tissue confirmation establishes whether it is a real subclonal alteration or a plasma artifact. The same archival block serves the PIK3CA, MAP2K1, and SDHA E350fs confirmations on one panel.

MAP2K1 P124S

Tumor-tissue NGS confirmation is essential before MEK-inhibitor consideration. P124 is a recognized MEK1 activating site (P124L/Q/S) with heterogeneous response data, but the variant is ctDNA-only here; tissue confirmation establishes whether it is clonal in the resected tumor or a low-allele-fraction subclonal event. Bundle it on the same block as the PIK3CA and SDHA E350fs confirmation.

POLE R1679C

The decision-relevant action is a cancer-predisposition genetic-counseling referral, not a therapy. R1679 lies outside the canonical POLE proofreading hotspots (P286R, V411L, S459F), and with TMB under 1 and MSS the polymerase-proofreading polyposis phenotype is not supported. Counseling lands the variant as a likely VUS and folds in the germline SDHA result so the family gets one coherent risk discussion. No specimen is needed; the counselor reviews the existing germline report.

KIT/PDGFRA wild-type

NF1 assessment is the relevant subtyping step (somatic NF1 sequencing with del/dup on the comprehensive tumor panel; germline NF1 is captured by the hereditary PGL/PCC panel). NF1-related GIST is the other KIT/PDGFRA-WT subtype that keeps SDHB staining intact, so it sits in this case's differential and would give the MAP2K1 P124S call a mechanistic home. It is medium priority because NF1 reads out of the comprehensive tumor panel already being sent for the ctDNA-variant confirmation, so it adds no separate draw; it matters most if the SDHA workup fails to confirm SDH deficiency.

Where to order these assays

Assay	Provider	Decision gated	Contact
Germline SDHA full-gene sequencing + del/dup (hereditary panel including SDHA, SDHAF2, SDHB, SDHC, SDHD, MAX, TMEM127, VHL, RET, NF1)	**Labcorp Genetics (formerly Invitae) (preferred) (Invitae Hereditary Panel)**	**Confirms SDH-deficient driver call for belzutifan (NCT04924075) and temozolomide (NCT03556384/NCT05661643); triggers cascade testing of first-degree relatives.**	**test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037**
Germline SDHA full-gene sequencing + del/dup (hereditary panel including SDHA, SDHAF2, SDHB, SDHC, SDHD, MAX, TMEM127, VHL, RET, NF1)	GeneDx (GeneDx Panel)	Confirms SDH-deficient driver call for belzutifan (NCT04924075) and temozolomide (NCT03556384/NCT05661643); triggers cascade testing of first-degree relatives.	test info · 207 Perry Parkway, Gaithersburg, MD 20877 · 1-888-729-1206
Germline SDHA full-gene sequencing + del/dup (hereditary panel including SDHA, SDHAF2, SDHB, SDHC, SDHD, MAX, TMEM127, VHL, RET, NF1)	Ambry Genetics (PGLNext)	Confirms SDH-deficient driver call for belzutifan (NCT04924075) and temozolomide (NCT03556384/NCT05661643); triggers cascade testing of first-degree relatives.	test info · 15 Argonaut, Aliso Viejo, CA 92656 · 1-866-262-7943
Germline SDHA full-gene sequencing + del/dup (hereditary panel including SDHA, SDHAF2, SDHB, SDHC, SDHD, MAX, TMEM127, VHL, RET, NF1)	Mayo Clinic Laboratories (PGLPN / Hereditary Gene Panel)	Confirms SDH-deficient driver call for belzutifan (NCT04924075) and temozolomide (NCT03556384/NCT05661643); triggers cascade testing of first-degree relatives.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710

SDHB IHC re-cut with positive internal control (clone 21A11 or BSB-131), reviewed by a soft-tissue / GI pathologist familiar with SDH-deficient tumors	**Mayo Clinic Laboratories (preferred) (SDHB Immunostain, test code SDHB / 70550, technical component only)**	**Confirms dSDH-GIST phenotype that gates belzutifan (NCT04924075) and the temozolomide trials; resolves the SDHB-intact / SDHA-biallelic paradox before referral.**	**test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710**
SDHB IHC re-cut with positive internal control (clone 21A11 or BSB-131), reviewed by a soft-tissue / GI pathologist familiar with SDH-deficient tumors	Memorial Sloan Kettering Pathology Consultation Service	Confirms dSDH-GIST phenotype that gates belzutifan (NCT04924075) and the temozolomide trials; resolves the SDHB-intact / SDHA-biallelic paradox before referral.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-5511
SDHB IHC re-cut with positive internal control (clone 21A11 or BSB-131), reviewed by a soft-tissue / GI pathologist familiar with SDH-deficient tumors	Brigham and Women's Hospital Department of Pathology (Dana-Farber/BWH GIST referral)	Confirms dSDH-GIST phenotype that gates belzutifan (NCT04924075) and the temozolomide trials; resolves the SDHB-intact / SDHA-biallelic paradox before referral.	test info · 75 Francis Street, Boston, MA 02115 · 1-617-732-7510
SDHB IHC re-cut with positive internal control (clone 21A11 or BSB-131), reviewed by a soft-tissue / GI pathologist familiar with SDH-deficient tumors	NeoGenomics Laboratories (SDHB IHC)	Confirms dSDH-GIST phenotype that gates belzutifan (NCT04924075) and the temozolomide trials; resolves the SDHB-intact / SDHA-biallelic paradox before referral.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
SDHA IHC (clone 2E3 or D6J9M) on archival FFPE	**Mayo Clinic Laboratories (preferred) (SDHA Immunostain)**	**Arbitrates SDHA-vs-other-SDHx driver call; if SDHA IHC is lost it locks in dSDH-GIST regardless of SDHB result.**	**test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710**
SDHA IHC (clone 2E3 or D6J9M) on archival FFPE	Memorial Sloan Kettering Pathology Consultation Service	Arbitrates SDHA-vs-other-SDHx driver call; if SDHA IHC is lost it locks in dSDH-GIST regardless of SDHB result.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-5511
SDHA IHC (clone 2E3 or D6J9M) on archival FFPE	Brigham and Women's Hospital Department of Pathology	Arbitrates SDHA-vs-other-SDHx driver call; if SDHA IHC is lost it locks in dSDH-GIST regardless of SDHB result.	test info · 75 Francis Street, Boston, MA 02115 · 1-617-732-7510
SDHA IHC (clone 2E3 or D6J9M) on archival FFPE	NeoGenomics Laboratories (SDHA IHC)	Arbitrates SDHA-vs-other-SDHx driver call; if SDHA IHC is lost it locks in dSDH-GIST regardless of SDHB result.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Tumor tissue NGS confirmation of PIK3CA R93W (FoundationOne CDx, Tempus xT, or equivalent FDA-cleared comprehensive panel)	**Foundation Medicine (preferred) (FoundationOne CDx)**	**PI3Kα-inhibitor consideration (alpelisib, inavolisib); enrollment in PI3K-inhibitor trials.**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**

Tumor tissue NGS confirmation of PIK3CA R93W (FoundationOne CDx, Tempus xT, or equivalent FDA-cleared comprehensive panel)	Tempus Labs (Tempus xT CDx)	PI3Kα-inhibitor consideration (alpelisib, inavolisib); enrollment in PI3K-inhibitor trials.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Tumor tissue NGS confirmation of PIK3CA R93W (FoundationOne CDx, Tempus xT, or equivalent FDA-cleared comprehensive panel)	Caris Life Sciences (MI Cancer Seek / MI Tumor Seek Hybrid)	PI3Kα-inhibitor consideration (alpelisib, inavolisib); enrollment in PI3K-inhibitor trials.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
Tumor tissue NGS confirmation of PIK3CA R93W (FoundationOne CDx, Tempus xT, or equivalent FDA-cleared comprehensive panel)	NeoGenomics Laboratories (NeoTYPE Comprehensive Tumor Profile)	PI3Kα-inhibitor consideration (alpelisib, inavolisib); enrollment in PI3K-inhibitor trials.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Tumor tissue NGS confirmation of MAP2K1 P124S on a comprehensive panel (FoundationOne CDx, Tempus xT, MSK-IMPACT, or equivalent)	**Foundation Medicine (preferred) (FoundationOne CDx)**	**MEK-inhibitor consideration (trametinib, binimetinib) if the variant is clonal in tumor.**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
Tumor tissue NGS confirmation of MAP2K1 P124S on a comprehensive panel (FoundationOne CDx, Tempus xT, MSK-IMPACT, or equivalent)	Tempus Labs (Tempus xT CDx)	MEK-inhibitor consideration (trametinib, binimetinib) if the variant is clonal in tumor.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Tumor tissue NGS confirmation of MAP2K1 P124S on a comprehensive panel (FoundationOne CDx, Tempus xT, MSK-IMPACT, or equivalent)	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT)	MEK-inhibitor consideration (trametinib, binimetinib) if the variant is clonal in tumor.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Tumor tissue NGS confirmation of MAP2K1 P124S on a comprehensive panel (FoundationOne CDx, Tempus xT, MSK-IMPACT, or equivalent)	Caris Life Sciences (MI Cancer Seek)	MEK-inhibitor consideration (trametinib, binimetinib) if the variant is clonal in tumor.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
Cancer-predisposition genetic counseling referral with POLE R1679C variant reclassification review	**NSGC Find-a-Genetic-Counselor Directory (preferred)**	**Polyposis surveillance decision (almost certainly no for R1679C VUS at TMB <1 / MSS) and family cascade-testing decision.**	**test info · 330 N Wabash Ave, Suite 2000, Chicago, IL 60611 · 1-312-321-6834**
Cancer-predisposition genetic counseling referral with POLE R1679C variant reclassification review	Dana-Farber Cancer Genetics and Prevention Program	Polyposis surveillance decision (almost certainly no for R1679C VUS at TMB <1 / MSS) and family cascade-testing decision.	test info · 450 Brookline Avenue, Boston, MA 02215 · 1-617-632-2178
Cancer-predisposition genetic counseling referral with POLE R1679C variant reclassification review	MD Anderson Clinical Cancer Genetics Program	Polyposis surveillance decision (almost certainly no for R1679C VUS at TMB <1 / MSS) and family cascade-testing decision.	test info · 1515 Holcombe Boulevard, Houston, TX 77030 · 1-877-632-6789

Cancer-predisposition genetic counseling referral with POLE R1679C variant reclassification review	Memorial Sloan Kettering Clinical Genetics Service	Polyposis surveillance decision (almost certainly no for R1679C VUS at TMB <1 / MSS) and family cascade-testing decision.	test info · 1275 York Avenue, New York, NY 10065 · 1-646-888-4050
Tumor tissue NGS for SDHA E350fs (same comprehensive panel as the PIK3CA/MAP2K1 confirmation)	**Foundation Medicine (preferred) (FoundationOne CDx)**	**Clonal-architecture mapping; no direct intervention gating since biallelic SDHA inactivation already established.**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
Tumor tissue NGS for SDHA E350fs (same comprehensive panel as the PIK3CA/MAP2K1 confirmation)	Tempus Labs (Tempus xT CDx)	Clonal-architecture mapping; no direct intervention gating since biallelic SDHA inactivation already established.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Tumor-informed ctDNA MRD assay (Signatera) or tumor-naive comprehensive ctDNA panel (Guardant360, FoundationOne Liquid CDx)	**Natera (preferred) (Signatera, tumor-informed ctDNA MRD)**	**Recurrence-surveillance signal; complements imaging but does not by itself trigger therapy initiation in dSDH-GIST.**	**test info · 13011 McCallen Pass, Building A, Austin, TX 78753 · 1-650-249-9090**
Tumor-informed ctDNA MRD assay (Signatera) or tumor-naive comprehensive ctDNA panel (Guardant360, FoundationOne Liquid CDx)	Guardant Health (Guardant360, tumor-naive comprehensive ctDNA)	Recurrence-surveillance signal; complements imaging but does not by itself trigger therapy initiation in dSDH-GIST.	test info · 505 Penobscot Drive, Redwood City, CA 94063 · 1-855-698-8887
Tumor-informed ctDNA MRD assay (Signatera) or tumor-naive comprehensive ctDNA panel (Guardant360, FoundationOne Liquid CDx)	Foundation Medicine (FoundationOne Liquid CDx)	Recurrence-surveillance signal; complements imaging but does not by itself trigger therapy initiation in dSDH-GIST.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor-informed ctDNA MRD assay (Signatera) or tumor-naive comprehensive ctDNA panel (Guardant360, FoundationOne Liquid CDx)	Tempus Labs (Tempus xF / xF+)	Recurrence-surveillance signal; complements imaging but does not by itself trigger therapy initiation in dSDH-GIST.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Baseline **surveillance: rapid-sequence whole-body MRI (skull base to pelvis) plus Ga-68 DOTATATE PET/CT**	**Treating-center academic radiology (NCI-designated center with a neuroendocrine-tumor program) (preferred)**	**surveillance frequency and modality choice for an SDHx-deficient patient.**	**varies by region**
Baseline surveillance: rapid-sequence whole-body MRI (skull base to pelvis) plus Ga-68 DOTATATE PET/CT	RadNet (national imaging network)	surveillance frequency and modality choice for an SDHx-deficient patient.	test info · 1510 Cotner Avenue, Los Angeles, CA 90025 · 1-310-478-7808
Baseline surveillance: rapid-sequence whole-body MRI (skull base to pelvis) plus Ga-68 DOTATATE PET/CT	NIH Clinical Center (research-grade PGL imaging via the Pacak laboratory)	surveillance frequency and modality choice for an SDHx-deficient patient.	test info · 10 Center Drive, Bethesda, MD 20892 · 1-800-411-1222

Plasma free metanephrines and normetanephrines (LC-MS/MS) and/or 24-hour urine fractionated metanephrines and catecholamines	**Mayo Clinic Laboratories (preferred) (PMETF / Metanephrines, Fractionated, Free, Plasma)**	**Functional screen; informs anesthesia-safety planning and surveillance interval.**	**test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710**
Plasma free metanephrines and normetanephrines (LC-MS/MS) and/or 24-hour urine fractionated metanephrines and catecholamines	Quest Diagnostics (Metanephrines, Free, Plasma (LC-MS/MS))	Functional screen; informs anesthesia-safety planning and surveillance interval.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Plasma free metanephrines and normetanephrines (LC-MS/MS) and/or 24-hour urine fractionated metanephrines and catecholamines	Labcorp (Metanephrines, Fractionated, Plasma Free)	Functional screen; informs anesthesia-safety planning and surveillance interval.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
Plasma free metanephrines and normetanephrines (LC-MS/MS) and/or 24-hour urine fractionated metanephrines and catecholamines	ARUP Laboratories (Metanephrines, Fractionated, Free, Plasma by LC-MS/MS)	Functional screen; informs anesthesia-safety planning and surveillance interval.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-522-2787
SDHC promoter methylation analysis by pyrosequencing or methylation-specific PCR (research / academic referral assay)	**NIH Pediatric and Wild-Type GIST Clinic (preferred)**	**dSDH-GIST sub-driver call if SDHA inactivation is not confirmed; does not gate therapy by itself.**	**test info · 10 Center Drive, Bethesda, MD 20892 · 1-240-760-6000**
SDHC promoter methylation analysis by pyrosequencing or methylation-specific PCR (research / academic referral assay)	University Hospital Mannheim (Wegert / Hofmann group)	dSDH-GIST sub-driver call if SDHA inactivation is not confirmed; does not gate therapy by itself.	test info · Theodor-Kutzer-Ufer 1-3, 68167 Mannheim, Germany
MGMT promoter methylation by pyrosequencing or methylation-specific PCR	**Mayo Clinic Laboratories (preferred) (MGMTM / MGMT Promoter Methylation, Tumor)**	**Temozolomide-response prediction at recurrence (NCT03556384, NCT05661643).**	**test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710**
MGMT promoter methylation by pyrosequencing or methylation-specific PCR	Quest Diagnostics (MGMT Methylation Analysis)	Temozolomide-response prediction at recurrence (NCT03556384, NCT05661643).	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
MGMT promoter methylation by pyrosequencing or methylation-specific PCR	Labcorp (MGMT Promoter Methylation)	Temozolomide-response prediction at recurrence (NCT03556384, NCT05661643).	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
MGMT promoter methylation by pyrosequencing or methylation-specific PCR	NeoGenomics Laboratories (MGMT Methylation by PCR)	Temozolomide-response prediction at recurrence (NCT03556384, NCT05661643).	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
NF1 assessment: somatic NF1 sequencing + del/dup on the comprehensive tumor panel; germline NF1 covered by the row-1 hereditary PGL/PCC panel	**Foundation Medicine (preferred) (FoundationOne CDx)**	**WT-GIST subtype classification (SDH-deficient vs NF1-related) if SDH deficiency is not confirmed; informs the imatinib-sparing rationale.**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**

NF1 assessment: somatic NF1 sequencing + del/dup on the comprehensive tumor panel; germline NF1 covered by the row-1 hereditary PGL/PCC panel	Tempus Labs (Tempus xT)	WT-GIST subtype classification (SDH-deficient vs NF1-related) if SDH deficiency is not confirmed; informs the imatinib-sparing rationale.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
NF1 assessment: somatic NF1 sequencing + del/dup on the comprehensive tumor panel; germline NF1 covered by the row-1 hereditary PGL/PCC panel	Caris Life Sciences (MI Cancer Seek)	WT-GIST subtype classification (SDH-deficient vs NF1-related) if SDH deficiency is not confirmed; informs the imatinib-sparing rationale.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
NF1 assessment: somatic NF1 sequencing + del/dup on the comprehensive tumor panel; germline NF1 covered by the row-1 hereditary PGL/PCC panel	Labcorp Genetics (formerly Invitae) (Invitae Neurofibromatosis / germline NF1 analysis)	WT-GIST subtype classification (SDH-deficient vs NF1-related) if SDH deficiency is not confirmed; informs the imatinib-sparing rationale.	test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037
NF1 assessment: somatic NF1 sequencing + del/dup on the comprehensive tumor panel; germline NF1 covered by the row-1 hereditary PGL/PCC panel	NeoGenomics Laboratories (NeoTYPE Comprehensive Tumor Profile)	WT-GIST subtype classification (SDH-deficient vs NF1-related) if SDH deficiency is not confirmed; informs the imatinib-sparing rationale.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements
Germline SDHA full-gene sequencing + del/dup (hereditary panel including SDHA, SDHAF2, SDHB, SDHC, SDHD, MAX, TMEM127, VHL, RET, NF1)	Resolves whether SDHA L306P (VAF 44%) is germline (a single-hit carrier with a second somatic event) or fully somatic (biallelic somatic loss). The answer changes the inheritance counseling, dictates first-degree-relative cascade testing, and locks in the SDH-deficient driver call that gates the temozolomide trials (NCT03556384, NCT05661643) and the belzutifan wt-GIST cohort (NCT04924075). Tumor-only NGS that does not include matched germline cannot distinguish the two; SDHA full-gene sequencing with deletion/duplication analysis is the standard per the 2025 Florou review (pmid:39927693).	Labcorp Genetics (formerly Invitae) (Invitae Hereditary Panel) · test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037	5-10 mL whole blood (EDTA) or saliva kit

SDHB IHC re-cut with positive internal control (clone 21A11 or BSB-131), reviewed by a soft-tissue / GI pathologist familiar with SDH-deficient tumors	Biallelic SDHA inactivation (combined VAF ~85%) should destabilize the SDH complex and produce loss of SDHB staining; retained SDHB in this case conflicts with the molecular call. The most common explanation is a technical false-negative (under-fixed block, weak antigen retrieval, no internal-control vessel in the field), so a re-cut with an explicit positive internal control resolves it before the dSDH-GIST diagnosis anchors trial enrollment. If SDHB truly remains intact after a clean re-stain, the case becomes a outlier and the trial screener should flag it for the trial sponsors, since most dSDH-GIST trials use SDHB-IHC loss as the entry phenotype (pmid:35546442, pmid:23046294).	Mayo Clinic Laboratories (SDHB Immunostain, test code SDHB / 70550, technical component only) · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710	1 unstained slide from archival FFPE (block can return to pathology)
SDHA IHC (clone 2E3 or D6J9M) on archival FFPE	SDHA IHC is specifically lost in tumors with biallelic SDHA inactivation, while it is retained in tumors driven by SDHB/SDHC/SDHD or SDHC-promoter methylation (pmid:23023976, pmid:23459398). If this case is truly SDHA-driven, SDHA staining should be lost even when SDHB is read as intact; if SDHA also reads as intact after a clean stain, the molecular call must be reconsidered. SDHA IHC therefore arbitrates the SDHB-intact / SDHA-biallelic conflict more directly than re-staining SDHB alone.	Mayo Clinic Laboratories (SDHA Immunostain) · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710	1 unstained slide from archival FFPE

Tumor tissue NGS confirmation of PIK3CA R93W (FoundationOne CDx, Tempus xT, or equivalent FDA-cleared comprehensive panel)	R93W sits outside the validated alpelisib companion-diagnostic hotspots (E542K, E545K/A/D/G, H1047R/Y/L, C420R per the therascreen and FoundationOne CDx labels) and the ctDNA-only call has not been corroborated on tumor tissue. Confirming the variant in tumor DNA at a validated VAF establishes whether this is a real subclonal alteration or a ctDNA artifact, and frames any PI3Kα-inhibitor discussion. Without tissue confirmation the variant cannot be used to enroll in PI3K-inhibitor trials and clinical actionability stays uncertain (pmid:30880263, pmid:32663323).	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	1 FFPE block or 10-20 unstained slides; archival acceptable
Tumor tissue NGS confirmation of MAP2K1 P124S on a comprehensive panel (FoundationOne CDx, Tempus xT, MSK-IMPACT, or equivalent)	P124 is a recognized MEK1 activating site (P124L/Q/S) with documented but heterogeneous MEK-inhibitor response (pmid:25370473, pmid:25257211). The variant is currently ctDNA-only; tissue confirmation establishes whether it is clonal in the resected tumor or a low-allele-fraction subclonal/artifactual ctDNA event. Without tissue confirmation it cannot anchor MEK-inhibitor consideration, and MEK monotherapy in MAP2K1-mutant disease is already a stretch outside histiocytosis/melanoma without solid evidence of a clonal driver.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	1 FFPE block or 10-20 unstained slides; archival acceptable

genetic counseling referral with POLE R1679C variant reclassification review	R1679 sits outside the canonical POLE proofreading hotspots (P286R, V411L, S459F) and the tumor is TMB-low and MSS, so the standard polyposis phenotype is not supported here. The counseling visit puts the variant on the right side of the ledger (most likely VUS, no current ICI rationale, no polyposis surveillance trigger) and folds in the SDHA germline result once that returns so the family gets one coherent risk discussion rather than two staggered ones (pmid:32424176, pmid:32058550).	NSGC Find-a-Genetic-Counselor Directory · test info · 330 N Wabash Ave, Suite 2000, Chicago, IL 60611 · 1-312-321-6834	no specimen required; counselor reviews the existing germline report
Tumor tissue NGS for SDHA E350fs (same comprehensive panel as the PIK3CA/MAP2K1 confirmation)	The E350fs frameshift was called from ctDNA but not seen in the tumor NGS that already reported L306P and Y408N. Confirming or refuting E350fs in tissue clarifies whether it is a true third SDHA hit, a subclonal event present in a region not sampled by the original tumor block, or a ctDNA artifact. Decision impact is incremental rather than gating because biallelic inactivation is already established by L306P + Y408N; the value is in mapping clonal architecture before any future recurrence is biopsied.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	shared with PIK3CA/MAP2K1 block

Tumor-informed ctDNA MRD assay (Signatera) or tumor-naive comprehensive ctDNA panel (Guardant360, FoundationOne Liquid CDx)	The case is post-R0 with no measurable disease and the existing ctDNA call surface already showed SDHA E350fs plus PIK3CA and MAP2K1 variants, so plasma is informative in this patient. Serial ctDNA gives a lead-time signal for recurrence well ahead of cross-sectional imaging, and the assay can be either tumor-informed (Signatera; lower limit of detection but bespoke design takes 4-6 weeks) or tumor-naive (Guardant360 / FoundationOne Liquid CDx; off-the-shelf but less sensitive for low-tumor-fraction MRD). GIST has weaker prospective MRD validation than colorectal or lung, so frame this as informative rather than gating.	Natera (Signatera, tumor-informed ctDNA MRD) · test info · 13011 McCallen Pass, Building A, Austin, TX 78753 · 1-650-249-9090	5-10 mL plasma (Streck tube). Signatera baseline also needs an FFPE block.

Baseline surveillance: rapid-sequence whole-body MRI (skull base to pelvis) plus Ga-68 DOTATATE PET/CT	SDHx carriers carry a lifetime risk that warrants imaging surveillance even when the index tumor is a GIST; current consensus (Endocrine Society 2014, ENS@T) recommends biennial rapid-sequence whole-body MRI from skull base to pelvis, with Ga-68 DOTATATE PET/CT as the most sensitive functional tracer for SDHx-driven PGL (pmid:31369093, pmid:23934599). Establishing a clean baseline now also gives the recurrence-monitoring strategy a clean comparison point for the resected GIST. Whether to defer DOTATATE until the SDHA germline result returns is a reasonable cost discussion, but baseline whole-body MRI should not wait.	Treating-center academic radiology (NCI-designated center with a neuroendocrine-tumor program)	no specimen; imaging only
Plasma free metanephrines and normetanephrines (LC-MS/MS) and/or 24-hour urine fractionated metanephrines and catecholamines	An undetected functional paraganglioma in an SDHx carrier going to surgery or anesthesia for any reason is a hypertensive-crisis hazard, so biochemical screening pairs with the imaging surveillance as the standard SDHx workup (Endocrine Society 2014 guideline). Plasma free metanephrines by LC-MS/MS is the most sensitive single test; 24-hour urine fractionated metanephrines is the established alternative when plasma testing is unavailable.	Mayo Clinic Laboratories (PMETF / Metanephrines, Fractionated, Free, Plasma) · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710	10 mL plasma (EDTA) and/or 24-hour urine collection in acid-preservative container

SDHC promoter methylation analysis by pyrosequencing or PCR (research / academic referral assay)	Only relevant as a fallback if the SDHA story collapses on the SDHA IHC / germline / E350fs workup; in that scenario, SDHC promoter epimutation is the next-most-likely Carney-triad-pattern driver and most dSDH-GIST trials accept methylation-driven cases as SDH-deficient (pmid:31308404, pmid:25540324). Clinically-available testing is limited and the assay is mostly run at academic referral labs, so flag it now but defer the order until the SDHA workup returns.	NIH Pediatric and Wild-Type GIST Clinic · test info · 10 Center Drive, Bethesda, MD 20892 · 1-240-760-6000	FFPE block or 10 unstained slides
MGMT promoter methylation by pyrosequencing or PCR	Recent data show preferential MGMT promoter hypermethylation in SDH-deficient wild-type GIST and a corresponding rationale for temozolomide activity (pmid:38132569). If the temozolomide trials (NCT03556384, NCT05661643) become the leading systemic option at recurrence, MGMT status helps predict response and prioritize patients likely to benefit. The result does not gate enrollment but adds depth to the decision once disease becomes measurable.	Mayo Clinic Laboratories (MGMTM / MGMT Promoter Methylation, Tumor) · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710	FFPE block or 10 unstained slides; bundle with the SDHC methylation order if both are needed

NF1 assessment: somatic NF1 sequencing + del/dup on the comprehensive tumor panel; germline NF1 covered by the row-1 hereditary PGL/PCC panel	NF1-related GIST is the other KIT/PDGFRA-WT subtype that keeps SDHB staining intact, so it sits squarely in the differential raised by this case's retained-SDHB result. It is RAS/MAPK-driven through neurofibromin loss (pmid:16096406, pmid:27011036), which would also give the ctDNA MAP2K1 P124S call a coherent mechanistic home. Assessing NF1 matters mainly if the SDHA workup fails to confirm SDH deficiency; skipping it in that scenario would leave the WT-GIST subtype call, and the imatinib-sparing rationale, on weaker footing.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	shared FFPE block with the PIK3CA/MAP2K1 panel for somatic NF1; germline NF1 is included in the row-1 PGL blood/saliva panel

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.